



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 121: Structured Programming Language

Batch: 28th

Semester: Fall 2025

Date: 01/03/2026

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions.

5 × 8 = 40

1. (a) Define Array. Explain how arrays have simplified the declaration process and manipulation of data. CLO3 4
- (b) Write a C program that uses a two-dimensional array to store the marks obtained by students in different subjects and then calculates the average mark for each student. CLO4 4
2. (a) Convert between different Loop Structures in C – CLO4 2×2=4
 - i. Convert the following do-while loop into a for loop:


```
#include <stdio.h>
int main() {
    int i = 2, count = 0;
    while(count < 5) {
        printf("%d ", i);
        i += 2;
        count++;
    }
    return 0;
}
```
 - ii. Convert the following for loop into a while loop:


```
#include <stdio.h>
int main() {
    int n, sum = 0;
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        sum += i;
    }
    return 0;
}
```
- (b) Write a C program to print the following number pattern based on user input for the number of rows. The example pattern is when row = 4. CLO4 4


```
1
2 3
4 5 6
7 8 9 10
```
3. (a) Briefly explain the six elements of a function definition or function implementation in C with a general format. CLO4 4
- (b) Write two C functions demonstrating the following: CLO3 2×2=4
 - i. A function with no arguments and no return value &
 - ii. A function with arguments and one return value CLO4

4. (a) Explain the difference between call by value and call by reference with an example. CLO3 3
 (b) Define Recursive function. Write a C program that calculates the factorial of a given positive integer using recursion. The function should take an integer as input and return its factorial. CLO4 5

5. (a) Briefly describe the reference(&) and dereference(*) operators used with pointer variables in C with examples. CLO4 2
 (b) Find out the output of the following programs – CLO5 3×2=6

i. `#include<stdio.h>`
`int main()`
`{`
`int a = 6;`
`int *p = &a;`
`*p = *p * 2;`
`a = a + 4;`
`printf("%d %d", a, *p);`
`return 0;`
`}`

14, 16

ii. `#include<stdio.h>`
`int main()`
`{`
`int x = 5;`
`int *p = &x;`
`int **q = &p;`
`**q = **q + 3;`
`*p = *p * 2;`
`printf("x = %d\n", x);`
`return 0;`
`}`

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iii. `#include<stdio.h>`
`int main()`
`{`
`int x = 25, y = 10; // Assume address of x is 3000 and y is 4000`
`int *p, *q;`
`p = &x;`
`q = &y;`
`*p -= 5;`
`*q += *p;`
`p = q;`
`*p += 5;`
`x += 2;`
`printf("x = %d, y = %d, p = %d, q = %d\n", x, y, p, q);`
`return 0;`
`}`

37, 37, 4000, 4000

6. (a) State the differences between structure and union in C with their syntaxes. CLO4 3
 (b) Write a C program that defines a structure named Student to store a student's details, including ID(int), section(char) and mark(float). The program should take input for these details from the user using a loop and then display them. CLO4 5

7. (a) Explain the role of null character in character arrays. CLO3 2
 (b) Write a C program that compares two strings to check which one is lexicographically greater. CLO4 3
 (c) Determine the output of the following program snippets: CLO5 3

i. `int i;`
`for(i = 1; i <= 5; i++)`
`{`
`printf("%d ", i * 2);`
`}`

26

ii. `int i = 0, j = 0;`
`while (i < 5 && j < 10)`
`{`
`i++;`
`j++;`
`}`
`printf("%d %d", i, j);`

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HAMDARD UNIVERSITY BANGLADESH

Faculty of Science and Engineering
Department of Computer Science and Engineering

BSC (Honors) in CSE
ENG 127: English Skills

Final Examination

Fall 2025

Batch: 28th
March 03, 2026

2nd Semester
Time: 2 Hours

Total Marks: 40

CLOs	Question/ Statement	Level	Marks
CLO4	1. Some people believe that technology has made our lives more complicated, while others think it has made life easier. Discuss both views and give your own opinion.	Create	15×1=15
CLO4	2. Write a book review of any book that you have recently read. You should include: a. Name and genre of the book. b. A summary of the book. c. Recommendation.	Create	07×1=07
CLO4	Write a formal letter to the Registrar of your university requesting him to waive a penalty that you received for attendance issue.	Create	08×1=08
CLO2	4. Paraphrase the following text. In recent years, computer science has become an essential discipline that influences almost every aspect of modern life. From communication systems to healthcare services, software-based solutions are used to improve efficiency, accuracy, and accessibility. As a result, computer engineers are expected not only to possess strong technical skills but also to think critically and ethically while designing systems. With the rapid growth of artificial intelligence and data-driven technologies, the responsibility of computer professionals has increased significantly. They must ensure that technological advancements serve humanity without compromising privacy, security, or social values.	Understand	05×1=05
CLO1	What is a formal letter? Briefly discuss the types of formal letters are there?	Define	05×1=05



Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE127: Electronic Devices and Circuits

Batch: 28th

Date: 5-3-2026

Final Examination

Semester: Fall 2025

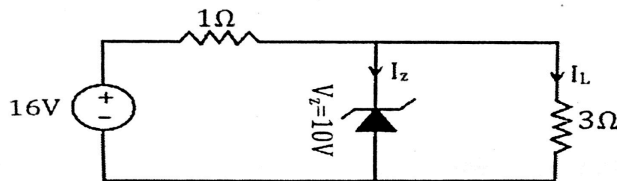
Time: 2 Hours

Total Marks: 40

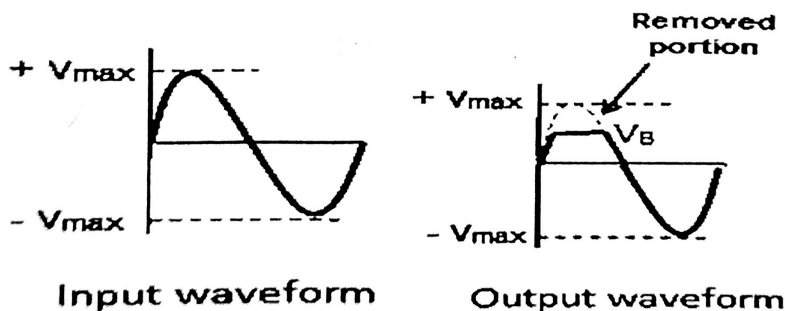
Answer any five questions of the followings.

$5 \times 8 = 40$

1. a) Mention and explain the types of breakdown related to diodes. CLO1 4
b) Describe the working principle of zener diode. How can a zener diode withstand breakdown? CLO1 4
2. a) Can a regular P-N junction diode withstand breakdown? Justify your answer with proper explanation. CLO1 3
b) Determine the value of I_Z from the following circuit : CLO2 5



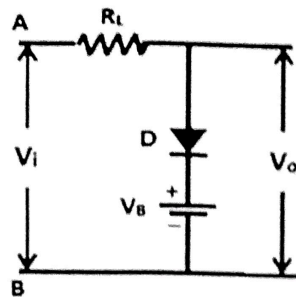
3. a) What is a clipper circuit? Classify diode clippers. CLO1 3
b) Can a series clipper and a parallel clipper provide the same output waveform? Justify your answer with proper example. CLO2 5
4. a) Design a clipper circuit for the given input and output waveform. CLO2 5



- b) If $V_i = 20\sin\theta$ and $V_b = 5V$, determine the output waveform of the given circuit assuming the diode is an ideal diode.

CLO2

3



5. a) Describe the working principle of a NPN transistor.

CLO1

4

- b) Find out the relation between α , β and γ in a transistor.

4

6. a) What are the available configurations for a transistor? Provide proper diagram for each configuration.

CLO1

2

- b) Explain common base configuration along with input and output characteristics.

CLO2

6

7. a) Describe the early effect and its effect on common emitter transistor configuration's input characteristics.

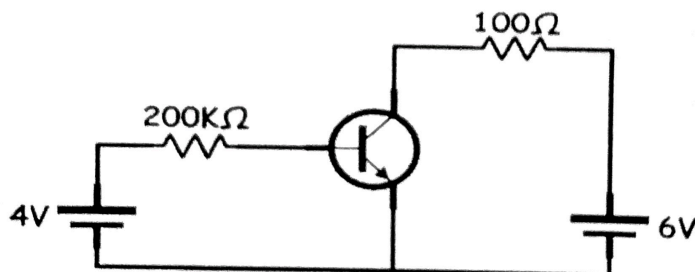
CLO1

4

- b) Find base current, collector current and output voltage in the given circuit assuming the transistor operates in active mode and $\beta = 50$.

CLO2

4





HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: PHY 121

Course Title: Physics II (Mechanics, Electricity, Magnetism and Modern Physics)

Final Examination

Date: 08/03/2026

Total Time: 2 Hours

Semester: Fall 2025

Total Marks: 40

Answer any three (04) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

		CLOs	Marks
Q1.	(a) Why is the magnetic force acting on a charged particle is always a sideways force?	CLO1	[2]
	(b) A current carrying conductor of a given length is placed in a magnetic field. Obtain an expression for the force exerted on the wire.	CLO2	[4]
	(c) At the equator, the earth's magnetic field is nearly horizontal, directed from southern to northern hemisphere. It's magnitude is about 0.50G. Find the force (magnitude and direction) on a 20m wire carrying a current of 30A parallel to the earth (a) from east to west, (b) from north to south.	CLO3	[4]
Q2.	(a) Define the torque in a magnetic field.	CLO1	[2]
	(b) Show that a current carrying loop immersed in a magnetic field is subjected to a torque which tends to rotate the loop around its axis. Hence obtain an expression for the torque acting on the loop.	CLO2	[5]
	(c) Analog voltmeters and ammeters, in which the reading is displayed by the deflection of a pointer over a scale, work by measuring the torque exerted by a magnetic field on a current loop. A spring provided a counter torque that balances the magnetic torque, resulting in a steady angular deflection ϕ corresponding to a given steady current i in the coil. The coil is 2.1 cm high and 1.2cm wide; it has 250 turns and is mounted so that it can rotate about its axis in a uniform radial magnetic field with $B=0.23T$. If a current of $100\mu A$ produces an angular deflection of 28° ($=0.49$ rad) what must be the torsional constant K of the spring?	CLO3	[3]
Q3.	(a) What is Hall effect?	CLO1	[2]
	(b) Define magnetic dipole and magnetic dipole moment.	CLO1	[2]
	(c) Explain the concepts of Hall voltage and Hall resistance with diagrams and equations.	CLO2	[4]
	(d) Explain the applications of the Hall effect.	CLO2	[2]
Q4.	(a) State Ampere's law.	CLO1	[2]
	(b) What is solenoid? What is a toroid?	CLO1	[2]
	(c) Apply Ampere's law to find the magnetic field at the Centre of a toroid.	CLO2	[3]

- (d) A solenoid is 1m long and 3cm in mean diameter. It has 5 layers of windings of 850 turns each and carries current of 5 Amp. (a) What is magnetic field B at its centre? (b) What is the magnetic flux ϕ_B for a cross-section of the solenoid at its centre? CLO3 [3]
- Q5. (a) State Biot-Savart law. CLO1 [1]
 (b) Apply Biot-Savart law to find the magnetic field due to a long straight wire carrying a current. CLO2 [6]
 (c) Two long and fixed parallel wires, A and B are 10cm apart in air and carry currents of 40 and 20 A, respectively in opposite directions. Determine the resultant flux density (a) on a line midway between the wires and parallel to them, and (b) on a line 8cm from wire A and 18cm from wire B. CLO3 [3]
- Q6. (a) State Maxwell's four equations in integral form. CLO1 [3]
 (b) What is motional electromotive force? CLO1 [1]
 (c) Explain the Lorentz transformation equations. CLO2 [6]



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT125 Course Title: Mathematics-II (Linear Algebra and Complex Variables)

Final Examination

Semester: Fall 2025

Date: 3/10/2026

Total Time: 2:00 Hours

Total Marks: 40

Answer any four (04) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No.	Questions	CLOs	Marks
Q1.	(a) What do you mean by analytic function?	CLO1	[1]
	(b) Prove that a necessary condition that a function $f(z) = u(x, y) + iv(x, y)$ be analytic over a region $\mathcal{R}$ is that the Cauchy-Riemann equations are satisfied in $\mathcal{R}$.	CLO2	[4]
	(c) Suppose that $u = x^2 - y^2$ and $v = -\frac{y}{x^2 + y^2}$. Prove that both u and v are harmonic but $u + iv$ is not analytic.	CLO3	[5]
Q2.	(a) What do you mean by basis and dimension of a vector space?	CLO1	[2]
	(b) Prove that, the vectors $\{(1, 2, 0), (0, 5, 7), (-1, 1, 3)\}$ form a basis for $\mathbb{R}^3$.	CLO3	[8]
Q3.	(a) Define harmonic functions.	CLO1	[2]
	(b) Suppose that $f(z) = u + iv$ is analytic in a region $\mathcal{R}$. Prove that u and v are harmonic in $\mathcal{R}$ if they have continuous second partial derivatives in $\mathcal{R}$.	CLO2	[3]
	(c) Prove that $u = e^{-x}(x \sin y - y \cos y)$ is harmonic and also determine v such that $f(z) = u + iv$ is analytic.	CLO3	[5]
Q4.	(a) Define limit and continuity of $f(z)$ at a point $z = z_0$.	CLO1	[2]
	(b) State and prove Cauchy integral formula.	CLO2	[6]
	(c) Prove that, $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$ does not exist.	CLO3	[2]
Q5.	(a) Define kernel and range of a linear transformation.	CLO1	[2]
	(b) If $T: U \rightarrow V$ be a linear transformation, prove that $\ker T$ is a subspace of U .	CLO2	[2]
	(c) Determine a basis for $\ker(T)$ and $\text{Im}(T)$ and also verify the dimension theorem for the linear transformation $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z, t) = (x - y + z + t, x + 2z - t, x + y + 3z - 3t)$	CLO3	[6]
Q6.	(a) What do you mean by residue?	CLO1	[2]
	(b) State and prove the Residue theorem.	CLO2	[3]
	(c) Prove that	CLO3	[5]

$$\int_{-\infty}^{\infty} \frac{z^2}{(z^2+1)^2(z^2+2z+2)} dz = \frac{7\pi}{50}$$